Computer Networks
Workshop No. 3 Subnetting

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INTRODUCTION

In the realm of modern enterprise networking, the design and implementation of scalable, secure, and efficient network architectures are critical to supporting organizational operations across geographically dispersed locations. This workshop submission outlines a methodical approach to subnetting and network design for a multi-location corporate entity, comprising ten distinct sites with heterogeneous operational requirements. The objective is to demonstrate proficiency in applying foundational subnetting principles while adhering to real-world constraints such as scalability, resource optimization, and security.

DEVELOPMENT

1. Network Design and Subnetting Plan

Common IP Address Range: 10.0.0.0/16 (Subnetted using VLSM for efficient allocation).

2. Location Specifications & Subnet Allocation Table

Location	Hosts (Current)	Growth (25%)	Total Hosts	Subnet Size	CIDR	IP Range	Subnet Mask
Headquarters (HQ)	500	125	625	1024	/22	10.0.0.0 – 10.0.3.255	255.255.252.0
Data Center (DC)	200	50	250	256	/24	10.0.4.0 – 10.0.4.255	255.255.255.0
Manufacturing Plant (MP)	150	38	188	256	/24	10.0.5.0 – 10.0.5.255	255.255.255.0
R&D Center (R&D)	100	25	125	128	/25	10.0.6.0 – 10.0.6.127	255.255.255.128
Branch Office 1 (BO1)	75	19	94	128	/25	10.0.6.128 – 10.0.6.255	255.255.255.128
Branch Office 2 (BO2)	60	15	75	128	/25	10.0.7.0 – 10.0.7.127	255.255.255.128
Sales Office (SO)	50	13	63	64	/26	10.0.7.128 – 10.0.7.191	255.255.255.192
Support Center (SC)	40	10	50	64	/26	10.0.7.192 – 10.0.7.255	255.255.255.192

Remote Office (RO)	30	8	38	64	/26	10.0.8.0 – 10.0.8.63	255.255.255.192
Warehouse (WH)	20	5	25	32	/27	10.0.8.64 – 10.0.8.95	255.255.255.224

Key Explanations

1. IP Range Choice:

- **10.0.0.0/16:** was selected for its large address space (65k+ hosts), enabling flexible subnetting across 10+ locations.

2. Subnetting Strategy:

- **VLSM** used to minimize IP waste. Subnets allocated in descending order of size (largest first) to prevent fragmentation.
- Growth factored into subnet size (e.g., HQ requires 625 hosts → allocated /22 subnet for 1024 hosts).

3. Security & Bandwidth:

- **Data Center** and **R&D** isolated with VLANs and firewalls.
- **HQ/DC:** 1 Gbps links; **Smaller offices:** 100–500 Mbps.

4. Assumptions:

- Growth capped at 25%.

Network Diagram Overview

A hierarchical WAN connects all locations to HQ and the Data Center. Subnets are routed through core switches, with edge routers managing inter-location traffic.